LS-CAT Management Board Meeting

Connection Instructions

To join the meeting on a computer or mobile phone:
https://bluejeans.com/8474912985

Phone Dial-in

312.216.0325 (US (Chicago))

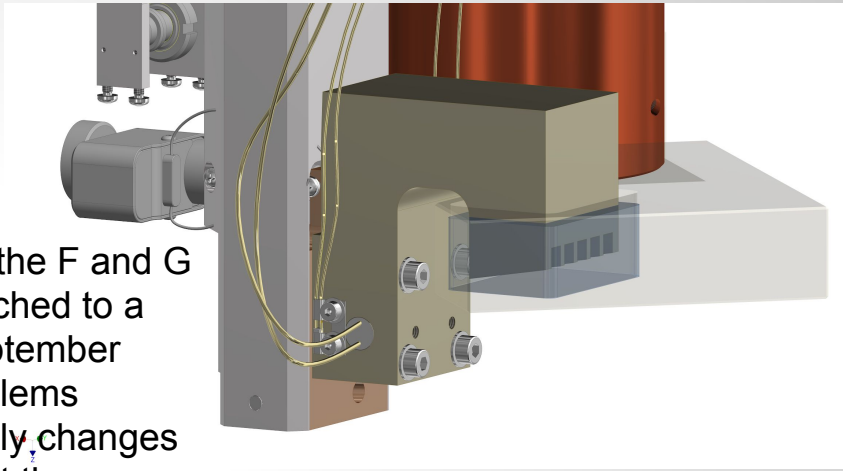
888.240.2560 (US Toll Free)

Meeting ID: 847 491 2985

Agenda

9:45	Sound Check / Informal Greetings
10:00	Welcome: Minutes and Agenda Approval (Stuckey)
10:10	LS-CAT Status Update (Brister)
10:25	Recent Review of LS-CAT by the APS SAC (Brister)
10:45	APS Users Meeting Workshop (Brunzelle)
10:55	Future Directions Committee Report (Forest)
10:05	APS Update (Streiffer)
11:25	New Business
11:30	Adjourn

LS-CAT Status



F-Mono Plan and Reality

We are (still) trying to resolve the water to vacuum braze in the F and G monochromators. The plan was to install an InGa bath attached to a properly made cooling block on both stations during the September down period. However, there were multiple installation problems leading to the installation being scrubbed. In the end the only changes were to re-plate the crystal holder with nickel and to remount the diamond on a film of InGa.

During start up we found the beam moved so much that it was practically unusable. Eventually the mono stabilized and the F station was put back into service in the last week of October.

In spite of the F mono down time (and the APS septum down time) the staff made sure that everyone who had sent crystals were served.

What happened? Likely this was caused by an excess of InGa being applied between the diamond and the cooling block.

D Mono Pains

- D Mono started having calibration problems (Nov 7)
- Kohzu (Japanese mono manufacturer) techs were on site and diagnosed an encoder signal problem
- Heidenhain (German encoder manufacturer) came and found 3 issues: 2 power supply voltages were low and the encoder itself was misaligned
- After these issues were addressed things have been running fine
- All told about a week of D time was lost
- All users for whom we received samples were served either on F, G, or on D after it came back on line (Thanks to the staff)

Budget Prediction for FY 2021

With Van Andel dropping out we'll lose a ¼ share of revenue putting a strain on our budget. At this point, however, we do not believe it will be necessary to increase the dues beyond the normal annual 3% but it may have an impact on our saved capital equipment fund if too many expensive things break.

LS-CAT in a Single Slide

2015 - 2019
(Last Review 2014)

Inputs

Institution:	GUP	NU	MSU	UM	UW	VU	UIUC	UIC	VARI	WSU
Allocation:	25%	15%	15%	15%	8%	8%	4%	4%	4%	4%
# People:	268	139	61	116	116	76	27	104	44	12

1. For each program we schedule in advance up to the stated allocation
2. Unscheduled allocations go into shared Rapid Access (RA) and advertised on our website
3. Programs with an unscheduled allocation may claim RA at any time
4. Programs with a full allocation may request RA one week ahead of time
5. When a user ends early the staff may offer the open station to LS-CAT users with a current ESAF
6. Unlike traditional 1980's style of beamline management, here Allocation ≠ Schedule ≠ Usage
7. Our Goals:
   - Maximize success of each user regardless of user program
   - Minimize unused beamtime

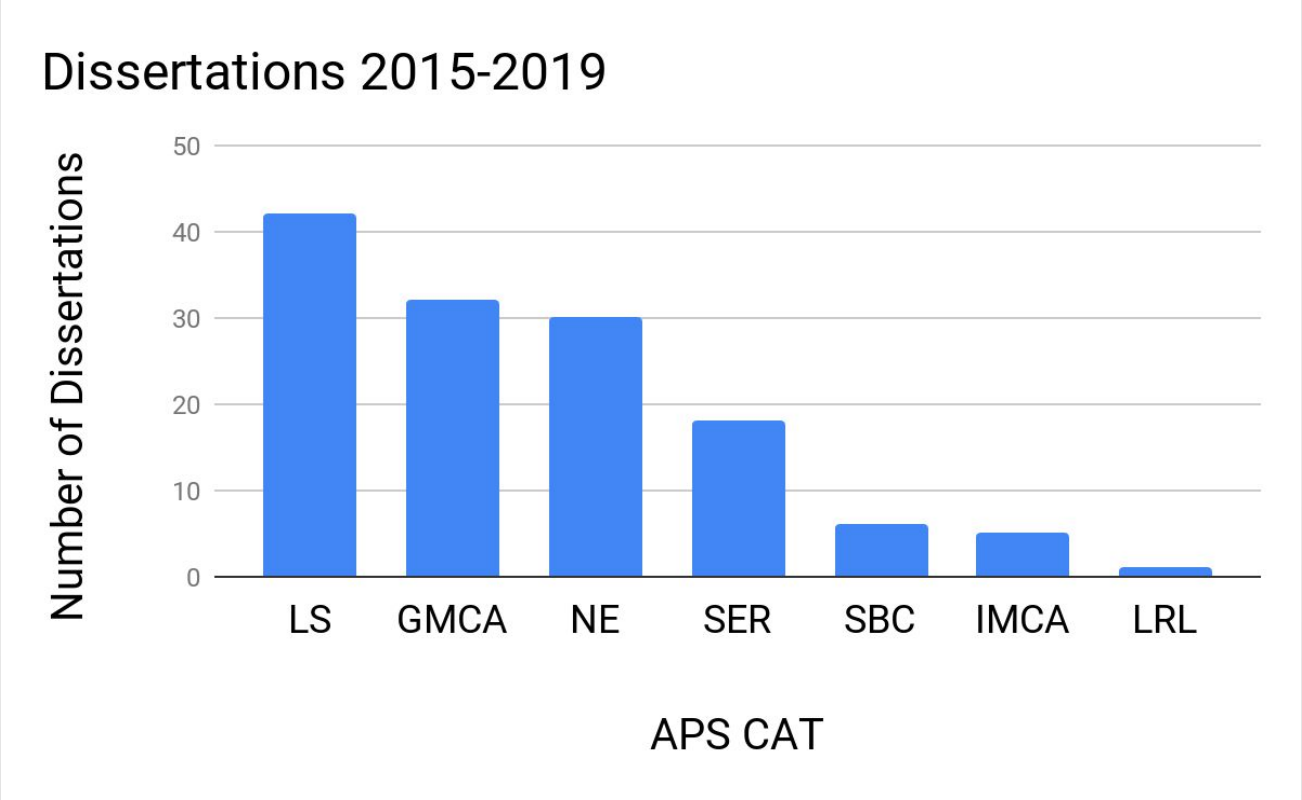
Outputs

- Journal Pubs 1331
- PDB Deposits 2155
- Theses 42
- 17% Industrial Experiments

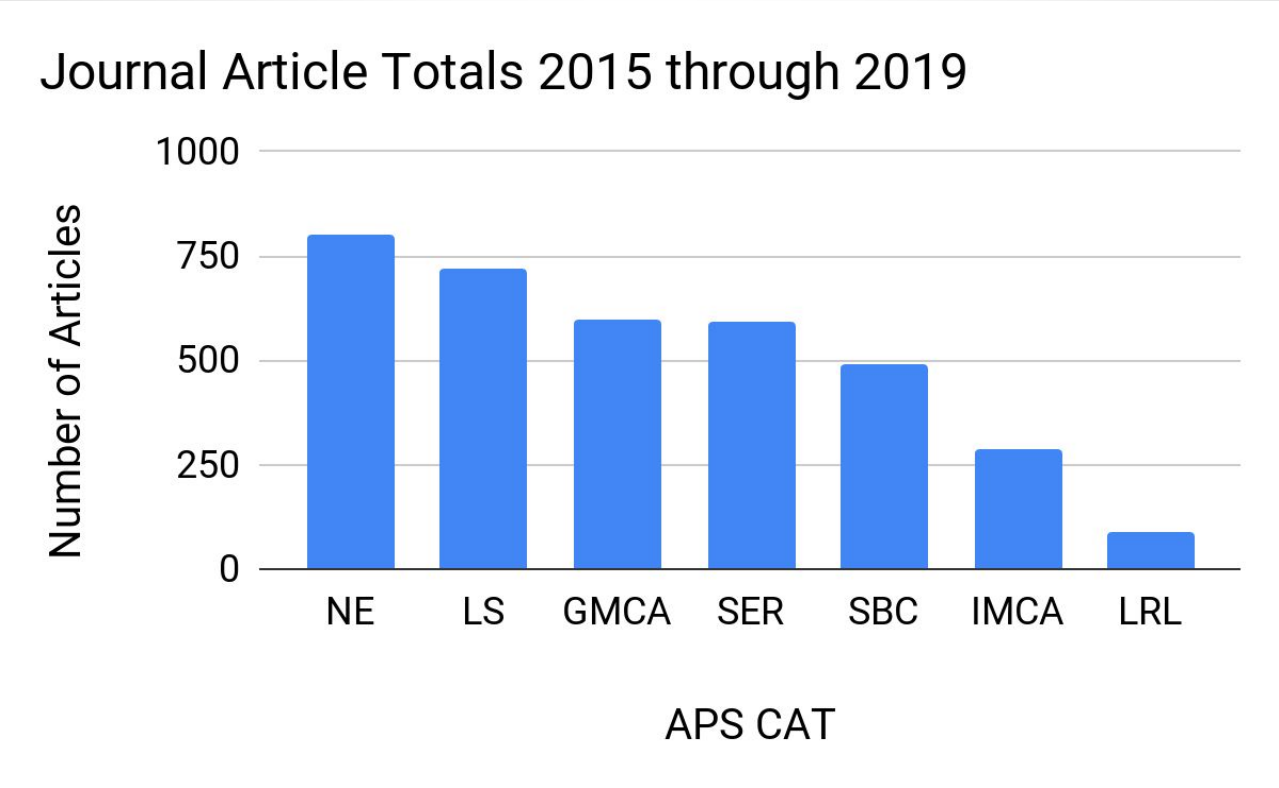
LS-CAT Review: User Program Statistics

User Program	People-Trips			Trips	PDB Deposits	Journal Articles
	On-Site	Remote	Mail-in			
NU	745	2	1	170	221	62
UIC	683	6	0	169	47	31
GUP	597	559	69	369	594	185
UW	291	403	2	163	122	78
UM	260	1761	3	666	236	133
UIUC	209	712	0	129	93	41
MSU	140	237	47	190	114	33
VU	46	753	4	288	171	88
WSU	8	801	163	237	19	21
VARI	2	356	2154	250	61	39
Total	2981	5590	2443	2631	1678	711
GUP %					35%	26%

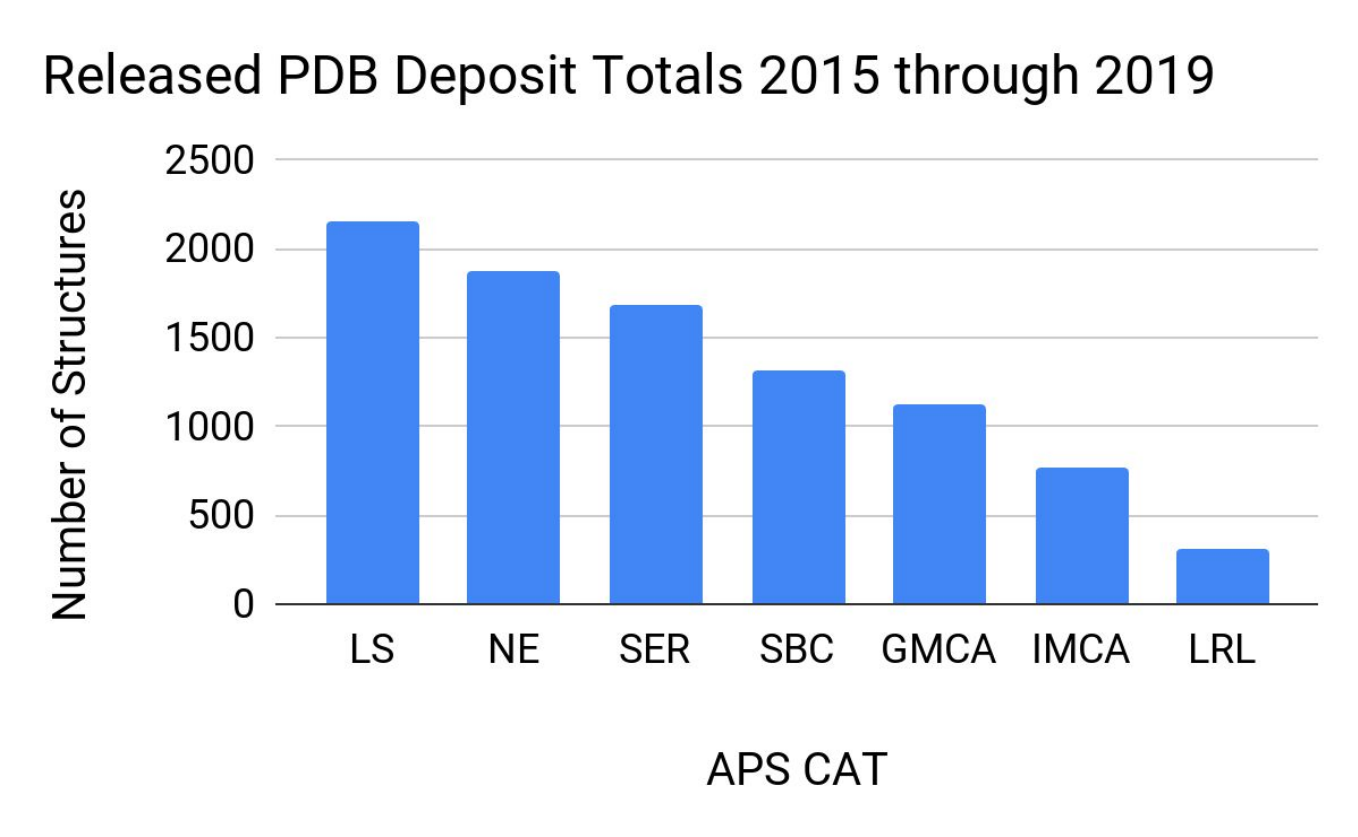
LS-CAT Review: Training the Next Generation



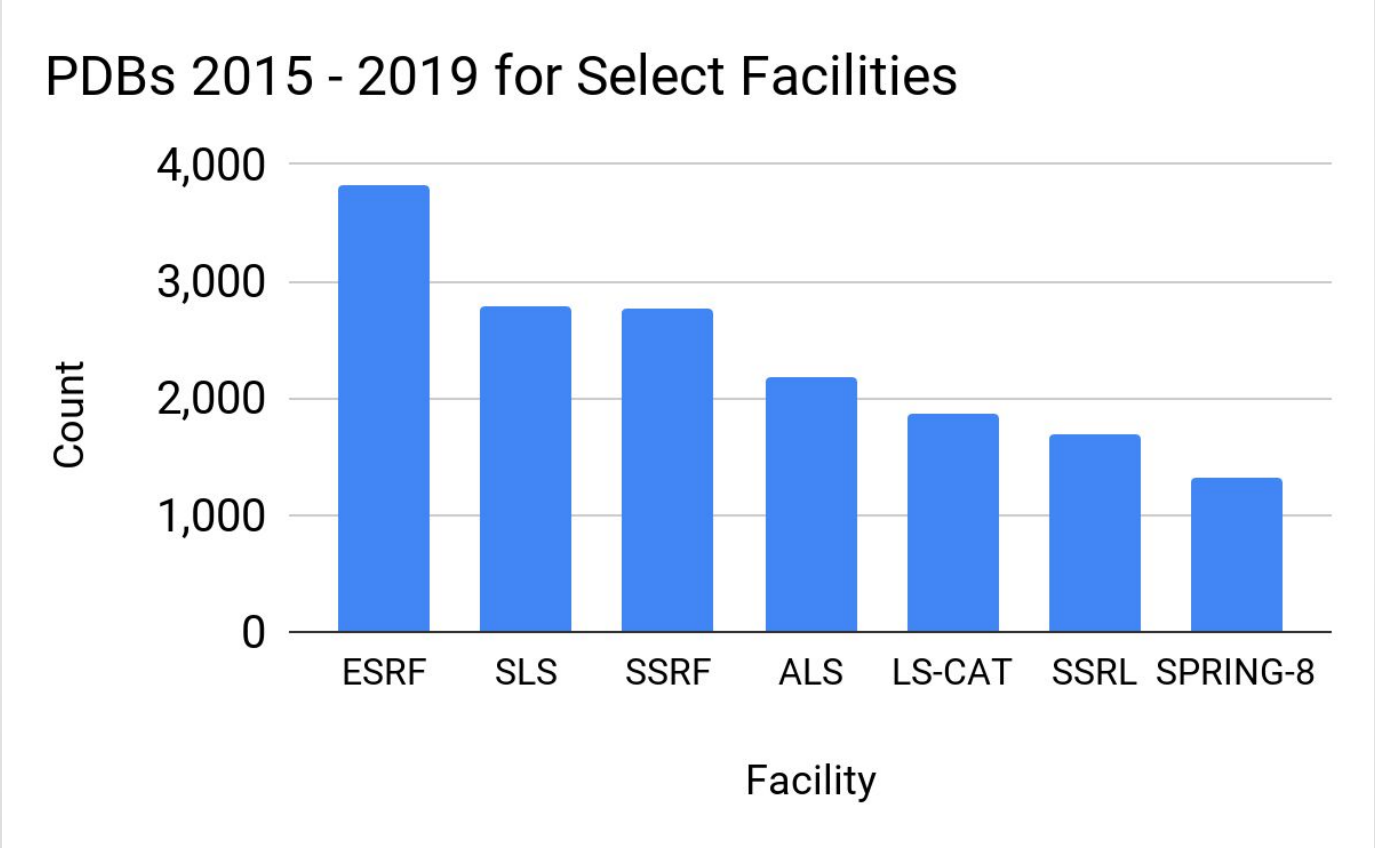
LS-CAT Review: Publications Compared to Peers



LS-CAT Review: Peer Comparisons



LS-CAT Review: World Wide Comparisons



LS-CAT Review Overall Rating

Overall rating for the beamline **Overall Rating: 1.5**

LS-CAT does an outstanding job of serving the MX user community. They have evolved to become a mature facility that is highly reliable, well-run, sustains high-throughput capabilities and contributes considerably to the national biological and biomedical enterprise. The management is particularly commended for their ability to recruit and retain outstanding and dedicated staff who are committed to the success of the beamline and its users. Most notable is the ability of the management team to identify challenges, and not overreact, but to strategically consider and carefully implement a wide range of potential solutions. It is these last characteristics that provide great confidence in the future success of LS-CAT.

(Highlight added for emphasis)

Proposed 1-Day Hands-On Workshop @ 2020 APS/CNM Users Meeting

Organizers:

Joseph S Brunzelle and Spencer M Anderson

Title:

Macromolecular Synchrotron Data Collection Workshop @ LS-CAT

Description:

LS-CAT will host an X-ray diffraction data collection, strategy, and processing workshop with both theory and hands-on components for macromolecular crystallography. The workshop will start with lectures on data collect theory and strategy, followed by hands-on crystal mounting, data collection, and data processing at the LS-CAT beamlines. The participants will work with the LS-CAT staff on test case projects. Participants are encouraged to bring their own specimens for data collection, and to bring old data for the data-reduction tutorials.

LS-CAT has 3 state-of-the-art Insertion Device beamlines available and equipped with both a Pixel Array Detector (PAD), Eiger 9M and two Rayonixy CCD detectors, MX300. Attendees will learn about the latest methods and strategies in data collection and a multitude of software packages to process their data. Students are split into several groups for the hands-on training sessions. Upon completing the workshop, attendees will have gained the practical experience collecting and processing data.